

Finite-Domain Two-Stream Plasma Instability

Introduction:

The simplest models of plasma typically assume an infinite domain to make the problem simpler. However, in the real world we use bounded domains in our fusion devices, and it is important to look at what changes to instability are made when we restrict our domain.

When two streams of charged particles move relative to one another, they generate an electric field. This electric field leads to rapidly growing perturbation in plasmas, which affects wave-particle interactions and how energy is transferred. In fusion devices, it is important to have as little instability as possible to maintain a standing wave structure.

Instead of assuming that we have infinite plane waves, I now want to put the plasma into a finite domain $x \in [0, L]$ and explore what happens.

In the infinite domain case, we assume that

$$\phi(x) \sim e^{ikx}$$

Which turns the equations we normally get into a local dispersion relation

$$\omega = \omega(k) \Rightarrow 1 - \frac{\omega_p^2}{(\omega - ku_0)^2} - \frac{\omega_p^2}{(\omega + ku_0)^2} = 0$$

For two stream plasma. When we explore the system in a bounded domain, we change from finding $\omega(k)$ to finding discrete eigenvalues ω_n and spatial eigenfunctions $\phi_n(x)$.

We start by finding the 1D dispersion relation for cold, two-stream plasma. We let the variable i go from 1 to 2 for each beam in the two-stream plasma. We assume the densities are equal, $n_1 = n_2 = \frac{n_0}{2}$. We assume the background velocities are equal and opposite, $u_1 = -u_2 = u_0$. The equilibrium electric field is zero because the uniform distribution of charge does not produce a net electric field. Additionally, we assume a collisionless plasma, where the collision frequency is small compared to the characteristic plasma frequency. We also assume there is no magnetic field. To find our differential equation, we look at the continuity, momentum, and Poisson's equations. We then convert them to 1D and linearize the equations, combining them to form the equation we want. [1]

Continuity:

$$\partial_t n_i + \nabla \cdot (n_i \mathbf{u}_i) = \partial_t n_i + \partial_x n_i u_i = 0$$

Momentum:

$$\partial_t \mathbf{u}_i + (\mathbf{u}_i \cdot \nabla) \mathbf{u}_i = -\frac{e}{m} \mathbf{E} \Rightarrow \partial_t u_i + u_i \partial_x u_i = -\frac{e}{m} E$$

Poisson's equation:

$$\nabla^2 \phi = \frac{d^2 \phi}{dx^2} = -\frac{e}{\epsilon_0} \sum_i n_i^1$$

Linearize and add small perturbations:

$$n_i = n_i^0 + n_i^1$$

$$u_i = u_i^0 + u_i^1$$

$$E = E^1 = -\nabla \phi \Rightarrow E^1 = -\partial_x \phi$$

We assume that there is a time independence ($e^{i\omega t}$), but we do not assume that the solution is in the form of plane waves in space. We can then plug this into the momentum equation to get

$$\partial_t (u_i^0 + u_i^1) + (u_i^0 + u_i^1) \partial_x (u_i^0 + u_i^1) = -\frac{e}{m} E^1 \Rightarrow$$

$$(-i\omega + u_i^0 \partial_x) u_i^1 = \frac{e}{m} \partial_x \phi$$

$$u_i^1 = \frac{e \cdot \partial_x \phi}{m(-i\omega + u_i^0 \partial_x)}$$

Similarly, we can solve the continuity equation to get

$$\partial_t (n_i^0 + n_i^1) + \partial_x [(n_i^0 + n_i^1)(u_i^0 + u_i^1)] = 0 \Rightarrow \partial_t n_i^1 + n_i^0 \partial_x u_i^1 + u_i^0 \partial_x n_i^1 = 0 \Rightarrow$$

$$(-i\omega + u_i^0 \partial_x) n_i^1 + n_i^0 \partial_x u_i^1 = 0$$

We can then substitute the value for velocity to get

$$(\omega - k u_i^0) n_i^1 + n_i^0 \partial_x \left\{ \frac{e \cdot \partial_x \phi}{m(-i\omega + u_i^0 \partial_x)} \right\} = 0$$

$$n_i^1 = -\frac{n_i^0 e}{m} \cdot \frac{\partial_{xx} \phi}{(-i\omega + u_i^0 \partial_x)^2}$$

We then substitute this into Poisson's Equation to get the differential equation:

$$\frac{d^2 \phi}{dx^2} = -\frac{e}{\epsilon_0} \sum_i \left\{ -\frac{n_i^0 e}{m} \cdot \frac{\partial_{xx} \phi}{(-i\omega + u_i^0 \partial_x)^2} \right\} \Rightarrow$$

$$\frac{d^2\phi}{dx^2} = \sum_i \frac{\omega_{pi}^2}{(-i\omega + u_i^0 \partial_x)^2} \frac{d^2\phi}{dx^2} \quad (1)$$

Numerical Methods:

To solve this differential equation, we want to introduce auxiliary variables:

$$g_i(x) \equiv (-i\omega + u_i^0 \partial_x)^{-1} \left(\frac{d^2\phi}{dx^2} \right)$$

$$(-i\omega + u_i^0 \partial_x)^{-1} g_i(x) = \frac{d^2\phi}{dx^2}$$

We then define

$$h_i(x) \equiv (-i\omega + u_i^0 \partial_x)^{-1} g_i(x)$$

This gives us a full, closed system of coupled differential equations:

$$\begin{cases} \phi''(x) = (-i\omega + u_i^0 \partial_x) g_i(x) \\ g_i(x) = (-i\omega + u_i^0 \partial_x) h_i(x) \\ \phi''(x) = \sum_i \omega_{pi}^2 h_i(x) \end{cases} \quad (2)$$

To solve this numerically, we want to convert this into a matrix form with the independent variables ϕ, g_1, h_1, g_2, h_2 .

$$\begin{cases} u_i \partial_x g_i - \phi'' = i\omega g_i \\ u_i \partial_x h_i - g_i = i\omega h_i \\ \phi'' - \sum_i \omega_{pi}^2 h_i = 0 \end{cases}$$

We want to assemble this into a standard matrix form, where we want

$$AX = i\omega BX$$

$$X = (\phi, g_1, h_1, g_2, h_2)^T$$

Each component of X is discretized on a one-dimensional grid of N points. Thus, $X \in \mathbb{C}^{5N}$.

The left-hand side (A) contains special derivatives

$$\begin{aligned}
D^2\phi &= \sum_i \omega_{pi}^2 h_i \\
u_i D h_i - g_i &= i\omega h_i \\
u_i D g_i - D^2\phi &= i\omega g_i
\end{aligned}$$

Which yields the final A matrix value of

$$A = \begin{bmatrix} D^2 & 0 & -\omega_{p1}^2 I & 0 & -\omega_{p2}^2 I \\ -D^2 & u_1 D & 0 & 0 & 0 \\ 0 & -I & u_1 D & 0 & 0 \\ -D^2 & 0 & 0 & u_2 D & 0 \\ 0 & 0 & 0 & -I & u_2 D \end{bmatrix} \quad (3)$$

And a B matrix value of

$$B = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & I & 0 & 0 & 0 \\ 0 & 0 & I & 0 & 0 \\ 0 & 0 & 0 & I & 0 \\ 0 & 0 & 0 & 0 & I \end{bmatrix} \quad (4)$$

The matrices A and B are constructed as block matrices, where each block is a $N \times N$ operator that acts on one field component ($A, B \in \mathbb{C}^{5N \times 5N}$). Spatial derivatives are approximated by using second-order central finite differences. [2]

The first derivative operator (D) is defined as

$$(Df)_i = \frac{f_{i+1} - f_{i-1}}{2\Delta x}$$

The second derivative operator (D^2) is defined as

$$(D^2 f)_i = \frac{f_{i+1} - 2f_i + f_{i-1}}{(\Delta x)^2}$$

These operators are implemented in the code as sparse tridiagonal matrices.

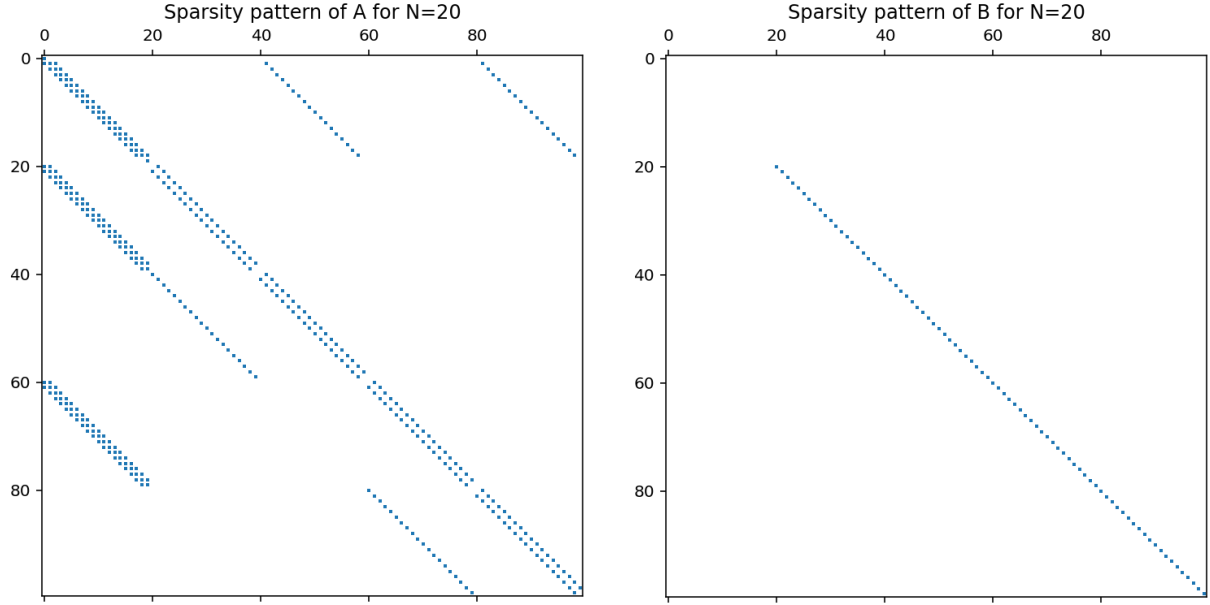


Figure 1. These plots show the sparsity pattern of A and B for $N = 20$

Both plots are broken up into a 5×5 system of blocks. In blocks where the matrices have an I term, there are only values on the diagonal. In blocks where the matrices have a D term, there are only values on the off diagonals. In blocks where the matrices have a D^2 term, there are values on the diagonal as well as the off diagonals.

These definitions break down at the boundaries, because we do not use ghost nodes or periodic boundary conditions. To solve this problem, we implement boundary conditions at either end of the domain.

$$\phi(0) = 0, \quad \phi(L) = 0 \quad (5)$$

In the matrix system, we implement the boundary conditions by replacing the corresponding rows of the global matrix system. The condition $\phi(0) = 0$ is implemented by replacing the first row of the matrix equation with

$$\phi_0 = 0 \rightarrow A_{0,:} = 0, \quad A_{0,0} = 0, \quad B_{0,:} = 0$$

Similarly

$$\phi_{N-1} = 0 \rightarrow A_{N-1,:} = 0, \quad A_{N-1,N-1} = 0, \quad B_{N-1,:} = 0$$

Results:

The numerical solution yields a discrete set of eigenvalues ω_n , and corresponding eigenfunctions $\phi_n(x)$, which represent the normal modes of the system in a finite domain.

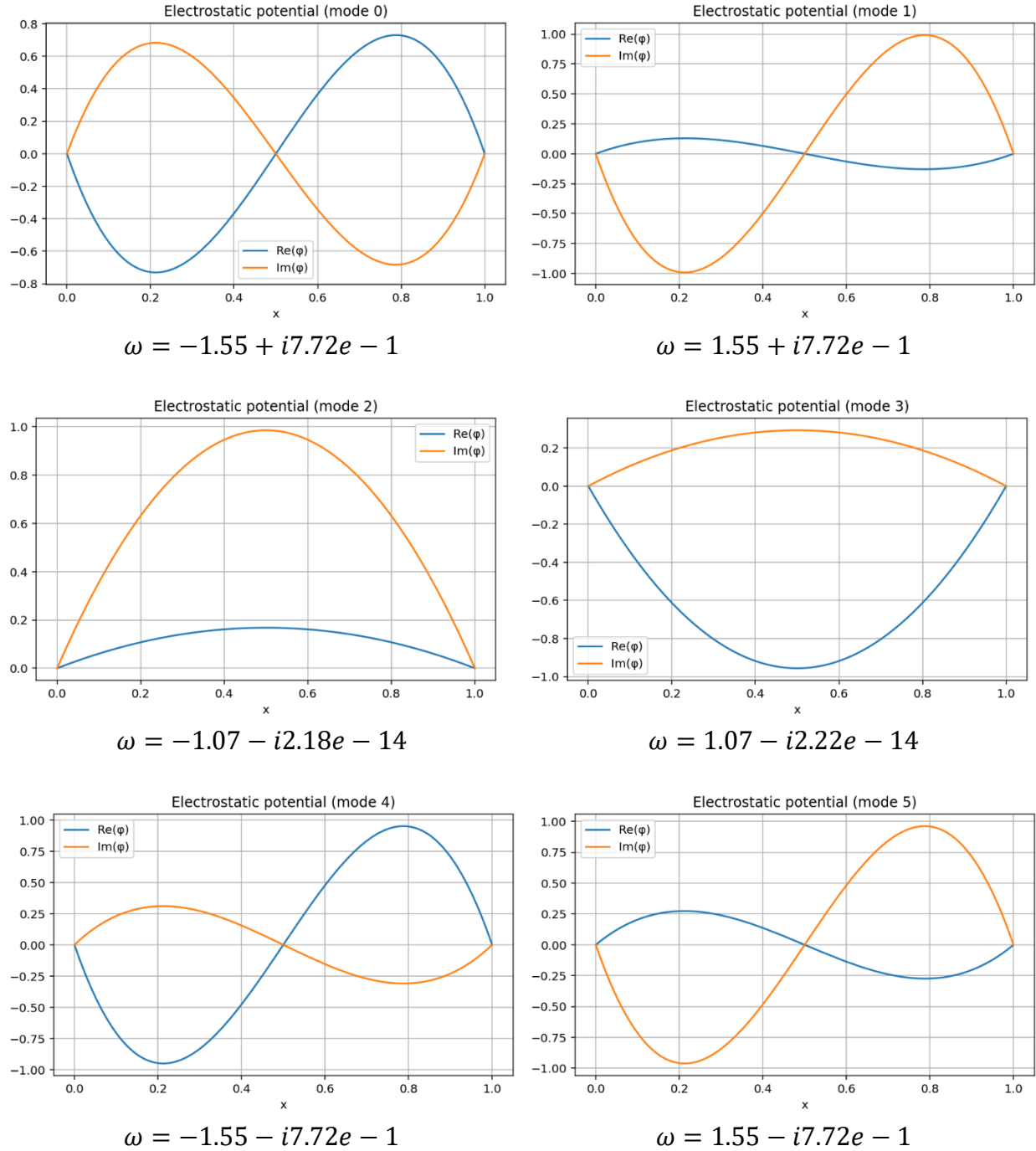


Figure 2. These plots show different eigenvectors of the potential for different eigenvalues. Eigenvalues with a negative imaginary part will exhibit instability growth.

The instability is determined not by the spatial structure of $\phi_n(x)$, but by the imaginary part of the corresponding eigenvalue

$$\omega = \omega_r + i\gamma_n$$

The modes that have $\gamma_n > 0$ will grow exponentially in time, and are thus unstable, while the modes that have $\gamma_n < 0$ are damped in time, making them stable.

The numerical solution only gives a subset of the eigenvalue spectrum of the discretized system. The full system contains $5N$ eigenvalues, but we only compute a few of them that are targeted near a specific shift, which was 0 for these plots. The observed eigenfunctions all contain the standing-wave structure, with either 0 or 1 interior zero. This is due to the fact that each mode corresponds to low spatial wavenumbers. When we move from the infinite domain to a finite one, the wavenumber moves from being a continuous value to being a discretized value, with approximate value of $k_n \sim \frac{n\pi}{L}$.

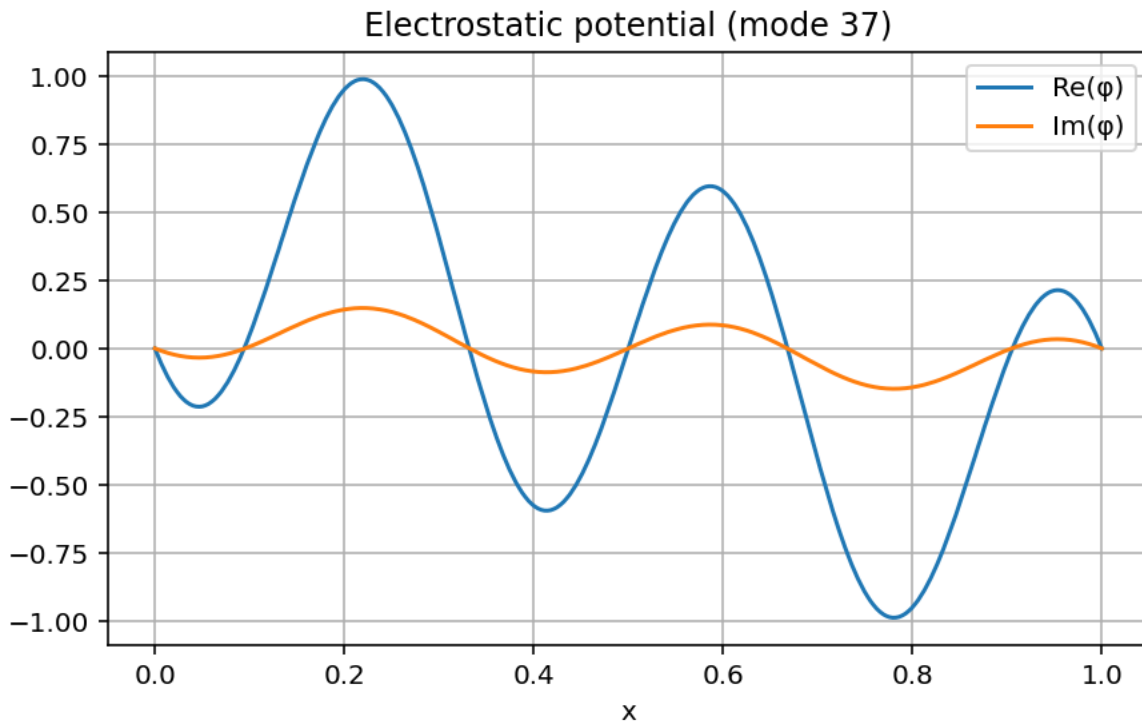


Figure 3. This plot shows potential for $k=50$, $\sigma=5$, and $\text{mode}=37$. $\omega = -17.15 - .932i$

By changing the shift that we look for as well as looking for more eigenvalues, the potential can attain many more zeros. However, the plots all display even or odd symmetry about $x = .5$. This is because in these plots, we assume that the background velocities are equal, $u_1 = -u_2$. If we change our code so that $|u_1| \neq |u_2|$, we get plots that aren't symmetric in the domain.

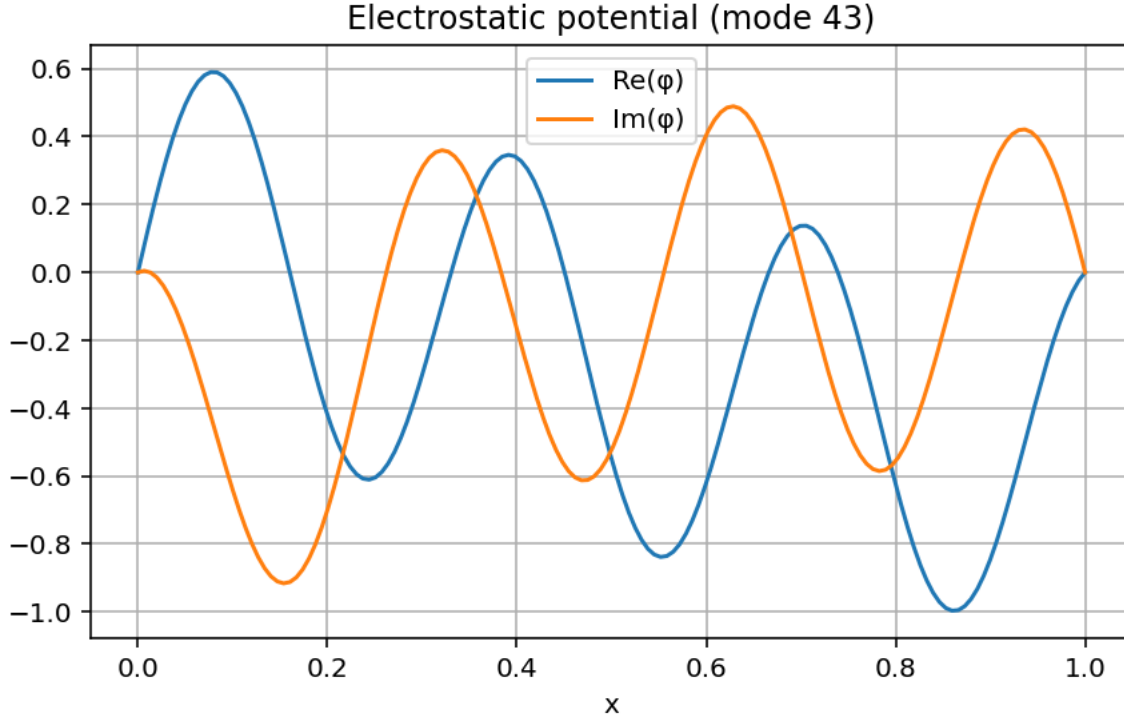


Figure 4. This shows a plot of the electric potential for $u_1 = 1$, $u_2 = -0.2$, $k=50$, $\sigma=5$, $\text{mode}=43$. $\omega = 4.06 - .99i$

Conclusions:

This project developed and implemented a numerical eigenvalue formulation of a modified two-stream plasma system in a finite one-dimensional domain. By removing the plane-wave assumption for potential, the governing equations were changed into a matrix eigenvalue problem of form $AX = i\omega BX$, where vector X contains the electrostatic potential and fluid response variables h_i and g_i discretized on a special grid. We approximated spatial derivatives by using second-order finite difference operators, and Dirichlet boundary conditions were enforced through direct modification of the A and B matrices.

Through this method, we were able to gather a discrete set of eigenmodes $\phi_n(x)$ with corresponding complex eigenvalues ω_n . The modes exhibited standing-wave structure consistent with the imposed finite-domain boundary conditions. A key difference between the finite domain and infinite domain is that the continuous spectrum of unstable wavenumbers found in the classical infinite-domain two-stream dispersion relation becomes discretized. The instability is no longer described by continuous $\omega(k)$ as in the dispersion relation, but rather by a discrete set of eigenvalues ω_n , where the imaginary part $\text{Im}(\omega_n)$ determines the growth or damping of each of the modes.

Computed eigenfunctions do not represent individual physical solutions to the problem but rather form a basis that can represent arbitrary perturbations in the system. Each of the modes responds in time proportionally to $e^{-i\omega_n t}$, and the overall plasma response can be expressed as a superposition of these modes.

There are significant limitations to the model and solution presented. In the mathematical model, we assume that there are only small perturbations, (hence the linearization step in our derivation) meaning that if there are any large or non-linear perturbations, the model will not reflect that accurately. Additionally, the accuracy of the numerical model is limited to second order in space, so there will be an error in anything higher order than that. Additionally, when we find the eigenvalues and eigenvectors, we only calculate a small proportion (typically 6) of the $5N$ (1500 for our case) total eigenvalues. The output is dependent of the chosen shift parameter, which biases eigenmodes toward specific frequency regions which we extract values from.

Potential next steps for this include comparing this numerical model to the dispersion relation we get from the classical two-stream dispersion relation in an infinite domain. We could map discrete eigenvalue to effective wavenumbers $k_n \sim \frac{n\pi}{L}$ and compare these values to the expected values of the two-stream dispersion relation. Another step to make is to alter the boundary conditions. For simplicity zero Dirichlet conditions were chosen, but that isn't physically accurate. In nature, there are boundary models using absorbing boundary layers and sheath-consistent models that better approximate the real solution.

References:

- [1] Introduction to Plasma Physics and Controlled Fusion, 3rd ed. (Springer, New York, 2016).
- [2] Numerical Analysis, 10th ed. (Cengage Learning, Boston, 2016).